

Ten Rack

A rack of six compact pedals - chorus / flanger, twin chorus, octave, drive, distortion and mod delay - in any order, in series, parallel or two split chains.



Overview

Ten Rack is a black 2U rack holding six compact pedals in the spirit of the mid-80s Japanese compact pedal series. The sound of each one is designed from scratch. There are six pedal types, and any slot can hold any of them - you can even use the same pedal twice, since every slot has its own memory and settings.

The six pedals are a bucket-brigade style chorus that morphs into a flanger (CF-1), a deep three-dimensional twin chorus (TW-2), an analogue-style octave divider (OC-3), a mid-hump overdrive with a built-in chorus or voice filter (SD-4), a high-gain distortion (DS-5) and a modulated delay (MD-6). Each has its own colour, and its knobs change to match the pedal in the slot.

The ROUTING keys choose how the six slots are wired: one long chain, all six side by side, or two chains of three. A path display above the pedals draws the signal flow, and changing a pedal or the routing crossfades smoothly, so there are no clicks. MASTER sets the dry / rack blend and the output level, and DRIFT lets the delay clocks and LFOs of every pedal wander like real analogue circuits.

Reach for it for

- Classic 80s clean chorus and echo on guitar and keys
- Drive, distortion and octave fuzz sounds built pedal by pedal
- Jet flanging and warped, pitch-bending echoes

- Lush, wide parallel pads from several modulation pedals at once
- Two complete rigs side by side, blended with one knob
- Thickening a bass or synth with the octave and drive

Controls

The header

SERIES / PARALLEL / SPLIT	The ROUTING. See Routing below. SERIES by default.
PRESETS	Opens the full preset list.
◀ / ▶	Step to the previous or next preset (wrapping round).
Preset display	Shows the preset number and name. Click it for the preset list too.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Signal path

The strip above the pedals shows how the slots are wired, from IN through the six slots to the MIX node and OUT. Each slot is drawn as a box with its number and pedal name, filled in its pedal's colour when it is on and as an outline when it is bypassed. The corner reads 1 > 6 (SERIES), 1 | 6 (PARALLEL) or A | B (SPLIT).

Routing

SERIES	One chain: 1 > 2 > 3 > 4 > 5 > 6, like a pedalboard. A bypassed slot passes the signal straight on.
PARALLEL	Every slot is fed from the input, and the pedals that are on are averaged at the output. With nothing on, the dry signal

passes through. Good for layering several modulation pedals without them feeding each other.

SPLIT

Two chains side by side: slots 1 - 3 make chain A and slots 4 - 6 chain B, each in series. A / B in MASTER blends the two. Each pedal shows an A or B next to its slot number.

A pedal slot

All six slots work the same way. The knobs change with the pedal in the slot (see The six pedals below).

Slot number	1 to 6, in a black circle at the top left. In SPLIT it is followed by A or B.
LED	Lit red while the pedal is on. Click it to switch the pedal on or off, like the footswitch.
◀ / ▶	Move the pedal one slot left or right. It swaps places with its neighbour, taking its knob settings and on / off state with it.
Knobs	Drag up or down (Shift for fine moves), or use the mouse wheel; double-click to reset to the pedal's default. While you touch a knob, its label shows the value.
Name plate	The pedal's name (CF-1 to MD-6) and what it is. Click it to choose a different pedal from the list; the new pedal starts at its own default settings. Drag it onto another slot to swap the two slots.
Footswitch	Switches the pedal on (ON) or off (BYPASS). A bypassed pedal dims a little. Switching crossfades over about 12 ms, and a pedal starts with empty delay lines each time it is switched on.

MASTER

A / B

SPLIT only (dimmed otherwise). Blends chain A and chain B, from A 100 (only A) through A = B (an equal mix, the default) to B 100 (only B).

MIX	The balance between the dry input and the rack, 0 - 100% (100% by default).
OUTPUT	Output level, -24 to +12 dB (0 dB by default).
TRIM	Input level before the rack, -24 to +12 dB.
GUARD	Click the TRIM knob (without turning it) to switch GUARD on or off; the small light at the knob's upper right shows it is on and glows brighter while it works. On by default. Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the pedals. It never boosts quiet sources.
SAT	Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises. 0% by default.
OUTPUT CLIP	A soft clipper on the output (always on) catches overs at the ceiling.
DRIFT	0 - 100% (25% by default). See Drift below.
IN / L / R meters	The input level and the left and right output levels.

Drift

DRIFT makes the clocks and LFOs of every slot wander slowly and independently, as on real analogue pedals. At 100% the LFO rates move by up to 25% either way and the bucket-brigade clocks (the delay times and the tone of the delay lines) by up to 3%, so two choruses never quite line up. At 0% everything is rock steady.

The six pedals

CF-1 CHORUS / FLANGER (lilac)

A bucket-brigade style delay swept by a rounded triangle LFO, with the dark, slightly soft top of the old delay chips. MORPH glides between a lush stereo chorus and a metallic flanger.

MORPH	0 - 100% (0% by default). Fully left is a wide stereo chorus (3 - 12 ms delay, the two sides sweeping in opposite directions); fully right a flanger (0.4 - 4 ms delay, a sweep of up to almost two octaves, with full REGEN). In between, it blends smoothly from one to the other.
RATE	The LFO speed, 0.1 - 10 Hz (35% = 0.5 Hz by default).
DEPTH	How far the LFO sweeps, 0 - 100% (55% by default).
MANUAL	Where the delay sits, shown in ms: 3 - 12 ms for the chorus, 0.4 - 4 ms for the flanger (50% by default). For the flanger this sets the pitch of the jet; for the chorus, how thick it sounds.
REGEN	Feedback, 0 - 100% (0% by default). It reaches up to 92% feedback at full MORPH for the ringing, whistling flange, but only about a third of that in chorus mode. The wet level trims down as REGEN rises to keep the level even.

TW-2 TWIN CHORUS (light blue)

Two delay lines per side, swept by two overlapping LFOs. The right side runs in antiphase, with a little cross-cancelling between the sides, for a deep, three-dimensional spread.

SPEED	The first LFO's speed, 0.08 - 4.8 Hz (40% = about 0.4 Hz by default).
DEPTH	How far the two lines sweep, 0 - 100% (60% by default).
RANGE	The second LFO's speed as a multiple of the first, x 1 to x 3.5 (x 2 by default). Higher settings give a busier, more animated shimmer.
WIDTH	The stereo spread, 0 - 100% (100% by default). At 0 both sides move together; at 100 they move in opposite directions.
LEVEL	The chorus level, 0 - 100% (50% by default). The dry signal dips a little as you turn it up.

OC-3 OCTAVE (dark blue)

An analogue-style octave divider. It cleans up the input, follows the pitch you play, and divides it down by one and two octaves. It works on the centre of the stereo image and puts the sub voices in the middle. Play single notes for the cleanest tracking (it follows roughly 35 Hz to 1.4 kHz).

OCT 1	The level of the voice one octave down, 0 - 100% (50% by default).
OCT 2	The level of the voice two octaves down, a square wave that follows your playing level, 0 - 100% (0% by default).
DIRECT	The level of your dry signal, 0 - 100% (100% by default).
GRIT	0 - 100% (20% by default). Low settings give a smooth, round sub; turning it up swaps in a square wave at your playing level plus some rectifier grit, for a buzzier, fuzzier octave.
TONE	A low-pass on the sub voices, 150 Hz - 4 kHz (45% = about 650 Hz by default). Lower for a deep, soft sub; higher for more bite.

A gate stops the divider chattering on silence.

SD-4 SESSION DRIVE (yellow)

A soft-clipping overdrive with the classic mid-hump: only the mids above about 720 Hz are clipped, while the clean signal rides along underneath, so the lows stay tight and the notes stay clear. It has two built-in extras, picked with MODE.

DRIVE	The gain into the clipper, 0 - 100% (45% by default), up to about 42 dB. The top end is rounded off more as the drive goes up, and the output is partly level-compensated.
LEVEL	The output level, -21 to +15 dB (50% = -3 dB by default).
TONE	A low-pass, 600 Hz - 12 kHz (50% = about 2.7 kHz by default).
AMOUNT	How much of the MODE extra is added, 0 - 100% (0% by default).

default, so the extra is off).

**CHORUS / VOICE
(MODE)**

CHORUS (the default) adds a short stereo mini-chorus to the drive. VOICE adds a resonant mid-voice filter at 850 Hz, for a vocal, singing lead.

DS-5 DISTORTION (orange)

A treble-boosted booster into a high-gain stage with warm, lopsided hard clipping, followed by the vintage blend tone stack: a low-pass and a high-pass mixed by TONE, with a mid scoop at noon.

DIST

The amount of distortion, 0 - 100% (50% by default).

LEVEL

The output level, -26 to +10 dB (50% = -8 dB by default).

TONE

0 - 100% (50% by default). Left is dark and thick, right is bright and cutting; in the middle the mids are scooped.

MD-6 MOD DELAY (green)

A delay with a modulated delay time. With a slow RATE and small DEPTH the repeats are gently chorused and pitch-warped; with more they are flanged and seasick. The feedback loop is darkened and saturates softly, and changing TIME glides like a tape machine, bending the pitch of the repeats.

TIME

The delay time, 20 ms - 1 s (73% = about 350 ms by default).

FEEDBACK

How many repeats, 0 - 95% feedback (35% by default). The repeats get darker and softer each time round.

RATE

The modulation speed, 0.05 - 8 Hz (50% = about 0.6 Hz by default).

DEPTH

The modulation depth, 0 - 9 ms (30% = about 0.8 ms by default). The two sides are modulated a quarter-cycle apart for a wide image.

LEVEL

The echo level, 0 - 100% (35% by default). The dry signal always stays at full level.

Presets

Every preset starts from Init and then changes a few pedals. Presets set the routing, A / B, MIX, DRIFT and all six slots; OUTPUT and the IN / OUT section are left as they are.

Init	The classic order in series - OC-3, SD-4, DS-5, CF-1, TW-2, MD-6 - every pedal at its defaults, with only the CF-1 chorus on.
80s Clean Chorus	A slow, deep CF-1 chorus into a short MD-6 slapback of about 200 ms, with a little more drift.
Crunch + Twin Chorus	SD-4 crunch into a wide TW-2 twin chorus and a quiet MD-6 echo.
Octave Fuzz	OC-3 with both octaves and plenty of GRIT into a hot DS-5 distortion.
Session Lead	SD-4 driven harder in VOICE mode for a singing mid-range, into a 375 ms MD-6 echo.
Shimmer Pad (parallel)	PARALLEL routing: a deep CF-1 chorus, a lush TW-2 and a long, modulated MD-6 side by side, with extra drift.
Dual Rig Split	SPLIT routing. Chain A: SD-4 with its mini-chorus, CF-1 chorus and a short echo. Chain B: DS-5, TW-2 and a longer echo.
Warped Echo	CF-1 a little past halfway to a flanger, into an MD-6 with deep modulation and plenty of feedback, with drift at 60%.
Jet Flanger	DS-5 distortion into the CF-1 at full flanger, with a slow sweep and high REGEN.
Sub Bass Synth	OC-3 with both octaves, high GRIT and a dark TONE, into a light SD-4 drive and a gentle chorus: a synth-like bass.

Tips

- Drive pedals before modulation and delay is the classic order; try a chorus or the octave before a drive for a wilder sound.
- OC-3 tracks best on clean single notes, so put it first in the chain, before any drive or chorus.
- For a doubling effect in SPLIT, put the same pedal in both chains with slightly different settings and set A / B to A = B.
- In PARALLEL the pedals are averaged, so each one sounds a little quieter than on its own; a bypassed pedal drops out of the average.
- Turn MD-6 TIME slowly while it is repeating for tape-style pitch bends.
- Set DRIFT to 0 for repeatable sweeps; a little more than the default makes two choruses in the rack sound less locked together.

File: gfx-ten-rack.jsfx